2026

COMPUTER PROGRAMING

ASSIGNMENT NO # 1

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REG NO: BF25NWELE0740

SECTION: ELECTRICAL POWER

SUBMITTED TO: SIR RIZWAN

Write a single C++ statement to accomplish each of the following:

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i. Declare the variables voltage, current, and resistance to be of type double.

```
double voltage, current,
resistance;
// Create three decimal-number variables to store voltage, current, and
resistance
```

---

ii. Display the message “Welcome to Object Oriented Programming in C++!” on the screen.

```
cout << "Welcome to Object Oriented Programming in
C++!";
// Show a welcome message on the screen for the user
```

---

iii. Read a value from the user and store it in an integer variable named age.

```
cin >>
age;
// Take age input from the keyboard and store it inside variable age
```

---

iv. If the variable temperature is greater than 100, print “Overheating detected!”.

```
if (temperature > 100) cout << "Overheating
detected!";
// Check temperature; if it crosses 100, warn about overheating
```

---

v. Compute the area of a circle with radius r and store it in area (π = 3.14159).

```
area = 3.14159 * r *
r;
// Calculate circle area using πr² and save the result in area
```

---

vi. Print the values of x, y, and z separated by a comma.

```
cout << x << "," << y << "," <<
z;
// Print x, y, and z neatly separated by commas
```

---

vii. Assign the sum of voltage and current to total.

```
total = voltage +
current;
// Add voltage and current, then store the result in total
```

---

viii. Display “Error: Division by zero” only if denominator equals zero.

```
 if (denominator == 0) cout << "Error: Division by
zero";
// Show error message only when denominator becomes zero
```

---

ix. Increment the variable count by 1 and then print its new value.

```
 cout <<
++count;
// Increase count by 1 first, then print the updated value
```

---

x. Read three floating-point numbers from the keyboard into a, b, and c.

```
 cin >> a >> b >>
c;
// Take three decimal numbers from user and store them in a, b, and c
```

---

xi. Print the total resistance when three resistors are connected in series. Please take the values of the resistances from the user.

```
cin >> R1 >> R2 >> R3, cout << "Total Resistance (Series) = " <<
(R1 + R2 + R3);
// Take three resistor values and add them because series resistances simply
add
```

---

xii. Print the total resistance when three resistors are connected in parallel. Please take the values of the resistances from the user.

```
cin >> R1 >> R2 >> R3, cout << "Total Resistance (Parallel) = "
<< 1 / ((1/R1) + (1/R2) + (1/R3));
// Take three resistances and apply parallel resistance formula
```

---

xiii. Print the value of the voltage across R2 (using the voltage division rule) when two resistors (R1 and R2) are connected in series. Please select the values of R1, R2, and the voltage yourself.

(Chosen values: R1 = 10Ω, R2 = 20Ω, V = 12V)

```
cout << "Voltage across R2 = " << 12 * (20.0 / (10 + 20));
// Use voltage division rule to calculate voltage across R2
```

---

xiv. Print the value of the current through R2 (using the current division rule) when two resistors (R1 and R2) are connected in parallel. Please select the values of R1, R2, and the voltage yourself.

(Chosen values: R1 = 10Ω, R2 = 20Ω, V = 12V)

```
cout << "Current through R2 = " << (12.0 / (10 + 20)) * (10.0 /
(10 + 20));
// Find total current then apply current division rule to get current through
R2
```

---

xv. Calculate the Semester GPA of your previous semester.

- **Enter all your subjects.**
- **Enter the credit hours.**
- **Calculate the semester GPA.**

```
cin >> credit >> grade, totalCredits += credit, totalPoints +=
credit * grade;
// Enter credit hours and grade points, then update totals
 cout << "Semester GPA = " << (totalPoints /
totalCredits);
// Divide total grade points by total credit hours to calculate GPA
```
